new random variable t. T is also random because it's a function of the random data, and so it will have some distribution, and we call that the sampling distribution of T.

But it's just the distribution of a random variable, in this case that random variable is T, a function of our random data X.

We looked at some examples.

We talked about properties of these sampling distributions, but they describe the variability of a statistic or a transformation of the data, and specifically we often work with the standard deviation of a sampling distribution, and we call that the standard error, the standard error of a statistic T or an estimator.

And as the sample size and increases, for many statistics, many sampling distributions, they will approach normality, and that's a result of the central limit theorem of stat 131.

So we looked at a couple examples. We talked about normal data or sampling IID from a normal population and in that case mu was unknown that was the parameter of interest but the variance sigma squared was known and we saw that the sampling distribution of the MLE for me the MLE of that mean that population mean was also normal centered around the true population mean mu but with this variance sigma squared over N so right this variance gets smaller and smaller as N gets bigger and bigger as we collect more and more data so we get more and and more precise.

We also looked at this uniform data example, which we had kind of already shown in the week before.

We saw that the MLE of this parameter theta, theta here, that MLE is the maximum observed X value, the maximum X.

And this was the sampling distribution, this order statistic for the maximum. This is the probability density function of that sampling distribution. And we also looked at its expectation, and we saw that was n over n plus 1 times theta. So it's a biased estimator, Its expectation is less than the true value theta, but it's consistent. This weight n over n plus one approaches one as we collect more and more data as our sample size goes to infinity. Then we ended just by visualizing a specific sampling distribution. This was a sampling distribution for the mean or sample proportion of these binary random variables. And so we did this for different sample sizes along the columns here, 5, 10, 30, 50, and then in each row for a different true population value. We saw a couple things. We saw as the sample size N increases going from left to right, this sampling distribution gets closer and closer normal. This last column all looks very close to normal. This first column, it doesn't really match the normal approximation very well, especially for this value of p close to the boundary. And we saw for p close to 0.1, again, close to the boundary there, the sampling distribution is more skewed and so it approaches normality more slowly so even though it approaches normality in all three of these cases it does so more slowly in this first row than in this last row. In this last row, the normal approximation is very good almost immediately. And then we saw as the sample size increases the sample distribution has lower and lower variance. This last column is more concentrated versus the first column more spread out more diffuse, more uncertainty there. So that was sampling distribution We're going to kind of take a detour today and talk about a new probability distribution called the kai square distribution. So a little bit more of a probability lecture today, not so much a statistics lecture, but it's a very useful distribution that's going to come up over the next few lectures. We're going to use it quite a bit. Today we're mainly going to introduce it and talk about its properties. You may have covered kai square distribution in stat 131, or maybe not. I don't teach it when I teach stat 131, so I'm going to cover it here. But as I said, we're going to use this kai square distribution a lot. It comes up in

a lot of different statistics applications, particularly when we're talking about variances or estimating variances. It'll come up when we test hypotheses, which we'll talk about in a couple weeks, and then when we evaluate goodness of fit, which we'll see is really just a specific type of a hypothesis test. So it comes up in different areas. We'll use it. Today we're just going to kind of introduce this new distribution. We'll do a couple examples working with it as well. well. So I'm going to define it here by just giving you the probability density function or the PDF.

So we say that a continuous random variable that I'll call X follows a kai square distribution. And this distribution, this kai square distribution, has one parameter.

And I'm going to call that parameter m. And it has a name. It's called the degrees of freedom. And that's just a name for the parameter. But this parameter M is the one parameter associated with the kai square distribution. and it has this probability density function. So the probability density function inputting x, the function of the parameter m looks like this, 1 over 2 to the m over 2 power. Our gamma function evaluated that m over 2. And then we have x to the m over 2 minus 1 times e to the negative x over 2. And the random variable has to be positive, so x is greater than 0 here. That's the probability density function for a kai squared distribution with m degrees of freedom, and we'll often denote it in this way. So this says x follows a kye square distribution with m degrees of freedom. This is just the Greek letter kai here. X follows a kai square distribution with m degrees of freedom, means it has this probability density. And if we look at this piece right here, x to some power times e to the negative x over two, does this kernel look familiar at all? Gamma, yeah. That's a gamma kernel, right? So we'll talk about why that is in a moment, but yes, it is a gamma kernel. Some important properties of the kai square distribution, helpful to know. First, it's mean or its expectation is just M. It's just that degrees of freedom parameter, M. So if you know M, you know the expectation automatically, it's the degrees of freedom. The variance is also straightforward. It's just two times M, two times our parameter. It has this nice addativity property. So if we have a random variable, I'll call x1, that follows a kai square distribution, and it has some degrees of freedom parameter, I'll call it m1, and we have another random variable, call x2, that's also kai squared distributed. And maybe it has some other degrees of freedom parameter, call it m2. And if these two random variables x1 and x2 are independent, and we take their sum, we add them together, that new random variable that we get by adding them together also follows a kye square distribution. distribution. And its degrees of freedom is just the sum of the degrees of freedom from the two original random variables. M1 plus M2 is the degrees of freedom. And vice versa, if we have an X3 that we get by summing X1 and X2, where each of those are kai squared with their own degrees of freedom, then X3 is kai squared with the sum of their degrees of freedom. Then that means the X1 and X2 must be independent. So we have this nice addativity property. If they're independent, we can add them together, get a new kye squared random variable. and just add the degrees of freedom accordingly. And then the last property here, the kai squared distribution is the sum of independent squared standard normal random variables. And so we often use z to denote a standard normal random variable. So if I have z1, z2, zm, that's just m different standard normal random variables that are independent, right? And if we take each of those standard normal random variables and we square them, so z1 squared, z2 squared, and so on, And we sum those squared standard normal random variables together. That gives us kai squared random variable with m degrees of freedom. One thing I'll mention, this parameter m, what we're calling the degrees of freedom, it doesn't necessarily have to be an integer. It could be any positive real number. But very often it will be an integer, and that's kind of the result of this last property here. We often get to these kai square random variables by adding together sums of independent normal random variables and so m will be a whole number here.

I'm going to jump ahead real quick to the end of this set of lecture notes.

I have this kai squared table here. Now when we're working with a kai squared

distribution, right, and we want things like probabilities.

We have the probability density function here, right? So we could integrate this to get those probabilities, but that's a really messy integral. We don't really want to do that by hand. Nobody really does that by hand. Nowadays you would do it with a computer software, saw examples of that in the reading, using R software to calculate probabilities. But we won't have access to our software for things like exams, and so instead we're going to go old school and use a table of probabilities. So this is a table of probabilities for the kai squared family of distributions. And I say family of distributions because it's for different values of M, a parameter, or degrees of freedom. And so specifically, what is this doing? it says if x has a kye square distribution m degrees of freedom this table gives the value of x so that's the probability that the random variable x the kai squared random variable is less than equal to input value little x is p or the pth quantile of the random variable i'm going to write that here probability that the random variable x is less than equal to little x in green here it's equal to p in blue so if i want to do a specific example maybe you for random variable x following kai square distribution five degrees of freedom that example I would look at this first column labeled m and I would find the appropriate parameter appropriate degrees of freedom in this case five so that tells me I'm looking in this fifth row here let this fifth row so I'm going to be looking in there if I circle this value 2.675 I circle that in green I look up above it in the very first row we see 0.25 here so we can match that up green green, well this is saying, probability with the random variable X is less than or equal to this green value, 2.675, is what? 5, the blue value.

2. So that's how we would read this table. Yeah. Represent like the sum of five normal random variables square.

It could. It could. Yeah. That is a way to generate a kai square random variable and very often it will represent that. It could just represent a random variable that follows a kai square distribution with five degrees of freedom. Yeah, very often, like we'll see. we get there by adding sums of independent standard normal and the variables. Yeah, so that's the context we'll typically work in. Like different from this? This table of like types of people who are... That doesn't sound familiar to me, so I don't think so. Yeah, I'd have to see exactly what it was to be sure. Also, one more thing to note, this table is two pages long, so you'll notice these probabilities on this page go from very small. 4, they keep going on the next page.

0.005 to 0. So 0.5 to very, very large, 0.995, the other end. So this is just a table showing the CDF, the cumulative distribution function, for a discrete set of values for this kai squared or family of kai squared distribution, because we have for different possible values.

Yeah, so it's a little hard to match up, but yeah, the first column is one, first column is one, and so on.

Second column is two. The PDF is not symmetric. Yeah, yeah, yeah, and we'll look at some plots in just a moment, yeah. The sum of normal distributions. It's the sum of squared normal distributions. And so squared standard normal random variables aren't necessarily symmetric. One other thing to note, so we are given from this table probabilities to the left of some value, but we know we can get probabilities to the right, probability that the random variable X greater than 2.675, for example, is just one minus the probability that it's less than or equal to. 25.

0.675, or 1 minus 0. So we're given probabilities to the left, but we can easily get probabilities to the right.

Yeah, we need to know M, so we know which specific kai square distribution we're working with, and then we need to know either a quantile, right? We need to either identify a column and then find the corresponding row, or we need to know

the value, like 2.675 here, and find the probability associated with it. Either one is fine, depending on the content. Yeah, you could try to interpolate.

I'm not going to. going to give you non-integer ms.

And so in practice, I would say use software to do it. But yeah, if you had to do it with a table, you could try to interpolate.

So that's the kai squared table. We can use it to get probabilities rather than integrating that probability density function. Here's what that probability density function kind of looks like for some different values of M, different values of that degrees of freedom parameter. So this first one is one degree of freedom. We see it's very heavily skewed. Most of the mass is to the left towards zero, but it has this kind of tail going out to the right. As we get bigger, you know, three, three, five, and so on, it gets kind of more bell-shaped. That skew kind of diminishes. And so as M increases, it becomes more and more symmetric, and it actually approaches a normal distribution. Again, that's a result of the central limit theorem. The central limit theorem, we remember, involves either sums or averages of a bunch of IID random variables. Well, we just saw we can view kai squared as a sum of a bunch of IID random variables. Yeah. At least for whole number ends it does. Yeah, it doesn't have a nice interpretation like that for non-integer ends. So it's not so much the skew, but the mean is increasing. And that's a result of this, right? The expectation is M. And we would expect if we're adding more things together, on average it'll get bigger. So what you're seeing is the mean is getting bigger. Here the mean is one.

Here the mean is 50. But the skew is becoming less and less. It's fairly symmetric here. It's not perfectly symmetric, but it's fairly symmetric. It's very much not symmetric in this first case. So the skew or the lack of symmetry goes down as we increase them, but the mean goes up.

I don't really like using thresholds like that. I think it's more of a gradual shift, right? Some people will say, you know, 30. can be a threshold, but certainly 31's better than 30, and 29 is not that far off from 30, so to me it's more of a gradual thing. Someone identified earlier, but there was a gamma kernel. It was a good observation, right? This is actually a special case of the gamma distribution. So the chi-square distribution is a gamma distribution. All kai-square distributions are gamma. Not all gama's are k-squared, so it's a special case. And specifically, it's a gamma where the alpha parameter in the gamma distribution is m over 2, and the beta parameter is just one half. So when we have one true parameter that can change, and that's M, right? We don't allow this beta parameter to change. It's always one half. But it is just a special case of the gamma distribution. Another result that I'll give you that can be helpful when working with kai square random variables. So say you have some kai squared random variable X, M degrees of freedom, and let's say we multiply it by some constant, constant C. So we create a new random variable, let's call it y, that we get by multiplying. this constant c with the random variable x, or x is kye square. Well, this new random variable y will also be gamma. It's no longer kai square, but it'll also be gamma, and it will be a gamma distribution where the first parameter, alpha is m over 2, and the second parameter, beta, is 1 over 2 times c, that constant that we used. So we can scale or multiply these kai square random variables, and we still remain in that gamma family. Yeah, there's a nice relationship between the two. the, you know, kye squared is a special case of gamma, and even if you scale a kai squared, it's still gamma.

It's also related to the normal distribution, right, which we kind of already saw. So suppose X, random variable X follows a standard normal distribution, means zero variance one.

Then if we square it, square that normal random variable, call that Y, that follows a kai squared distribution with one degree of freedom.

We have a standard normal random variable, we square it, we get a kai squared

one random variable, one degree of free. So this property, the squaring of a kai squared random variable giving us, or squaring of a standard normal random variable giving us kai squared 1, and that addativity property that says we can add up independent kai squared random variables and then just add up their degrees of freedom and remain kai squared. This gives us a way to generate or simulate from a kai squared distribution, given that we have a way to simulate standard normal random variables first. So let's say we have a way to generate these standard normal random variables, X1 through XM, the independent, so IID standard normal, and then we square all of them, add them together, so the sum of XI squared, that new thing will follow a kai squared distribution with m degrees of freedom. And here I'm just doing that exact thing.

So I'm doing this 10,000 times, 10,000 draws or simulated replicates from a kai square distribution. And specifically, I'm adding together three squared standard normal random variables when I do this. So for each of these 10,000 cases, I generate three independent standard normal random variables. I square them, and I add them together, right? Each of those should be a random draw from a kai square distribution with three degrees of freedom. So I do this 10,000 times. I'm going to plot a histogram of those simulated random variables, and I'm going to compare that to the true kye squared density with three degrees of freedom. And this is what this looks like. So the histogram is the actual observed or simulated values that I took. And then this red curve is the true probability density function for a kai squared random variable with three degrees of freedom. And we see they match up nearly perfectly. So you like repeatedly sampling like three values from a normal distribution like 10,000 times when you're just adding them all up and then? Squaring them and adding them. Yeah, so I can draw three independent standard normals, square them, add them up, put that in a bucket. Draw three more, square them, add them up, put that in a bucket. Do that 10,000 times. I have 10,000 things in my bucket, and this is the histogram of what's in that bucket. So it's a nice way to generate or simulate from this distribution, given that we have a way to generate standard normals first. Okay, let's do a couple applications or examples. Again, it's most useful in the lectures we'll talk about the next few lectures and hypothesis testing. We'll see some uses here as well. Okay, so we'll start with this variance estimation example. So let's say we have X1 through XN, and this is a random sample from a normal distribution with known mean, mu, but unknown variance, sigma squared. We've looked at a lot of examples where we didn't know the mean, but we knew the variance. This is kind of the opposite. We know the mean, but we don't know the variance. then the MLE for that variance is this. You actually did this as part of the last extra credit problem. It was one of the steps in that. But this is the MLE for that variance. We just take each random variable X, we subtract its mean or center it, and then we square those, add them together, divide by E. Or the average of the squared deviations from the mean. That is the MLE of the variance parameter, summa squared. Well, that's not the point of this question. The point is to find the distribution. of this thing, n times our MLE divided by the true parameter, sigma squared. And we want to find the distribution of that. So if we look at the numerator, N times sigma squared hat, n times this, that's just the sum of the squared deviations. The hat, sum, x minus the known mean mu squared. But we're doing n times the MLE divided by sigma squared. So we just need to divide that thing by sigma squared. N, sigma squared hat over sigma squared, sum are squared deviations from the mean xI minus mu squared divided by sigma squared now. The true but unknown variance. And this is the same sum of each xI minus mu divided by sigma squared. Now notice what's going on inside the parentheses here. We have a random variable x that's normally distributed with mean mu and variance sigma squared or standard deviation sigma. Taking that normal random variable, we're subtracting its mean, and we're dividing by its standard deviation. What will be the distribution of this thing in parentheses? Standard normal. We subtract its mean, we divide by a standard deviation. Each of these will be a standard normal random variable. So I could write this as sum, zi squared. Again, just using zi to represent a standard normal random variable. ZI, IID, normal, zero one, standard normal. So this thing that we want the distribution of, is literally the sum of n squared standard normal random variables. The sum of n squared standard normal random variables, kai squared. It's one of the properties we

had. The kai squared distribution is the sum of independent squared normal random variables. This thing, n, sigma squared hat over sigma squared, follows a kai squared distribution with n degrees of freedom. Sigma follows like a normal zero to one because you're just, like, it's a normal random variable and then you're just like shifting it. Yeah, if you have a normal random variable and you subtract its mean, its new mean is zero. And if you scale by one over the standard deviation, its new variance is one. So it's like when you learned about Z scores, we're literally just doing a Z score here. We center it and then we scale it. So we're removing the mean and scaling it to have variance one. Okay. Next example says, consider the same setup as the previous example. But now find the sampling distribution of the MLE itself, sigma squared a half. So we found the distribution n times the MLE divided by sigma squared, or the distribution of just the MLE. So a scaled version of this. Any ideas here? The sampling distribution of sigma squared high is like the scale. Oh, is it because you're multiplying by N and divided by a square? Yeah, if we took this and multiplied by sigma squared over N, those all cancel out and were just left with the MLE. So would it still be just k-squared? Is it kye squared if we scale a kye square random variable by a constant? Gamma, that's the idea. We're going to use this property here, the relation to the gamma distribution. We know if something is kai squared and we scale it by a constant C, the result will have this gamma distribution. We're just going to use that. That's the idea. Recall n sigma squared hat or sigma squared follows kai squared distribution, n degrees of freedom. That's what we found in the last problem. But that means that sigma squared hat, the MLE itself, is the result sigma squared over N times chi-squared random variable with n degrees of freedom. Take this, and we multiply by sigma squared over n, we get sigma-squared hat. In other words, sigma-squared hat, our M-LE of the variance, is equal to C times X, where X, k-squared random variable, n degrees of freedom. This constant C is just sigma-squared over n. This is kai squared n. We're multiplying it by sigma squared over n, and those cancel out giving us sigma squared hat. Okay, so now we can use that theorem, the relation to the gamma distribution. So a theorem 1.1 above, we know now the sigma squared hat, the MLE, follow a gamma distribution. The first parameter, alpha, is n over 2, and the second parameter is N over 2 sigma squared. The sampling distribution of the MLE of the variance. n over 2 sigma squared yeah the first one is in over 2 yeah one more thing i want to point out here wasn't part of the question but just note here expectation expectation sigma squared hat this m le because it's a gamma distribution we know the mean or the expectation of the gamma distribution is alpha over beta or over 2 divided by 2 sigma squared in this case and that's just sigma squared square our mle square hat is unbiased expectation is the true parameter the big property of distribution where if you have like, so you said n sigma squared over sigma squared that falls at high square distribution and it's like multiply both sides by sigma squared over n. As long as it's the constant, yeah, it's fixed, right? N and sigma squared are not random variables. So we're just scaling a random variable by a constant, just like in the setup of that theorem. If we were scaling it by something that was a random variable, all bets are off.

We can't do that. One more example, so the sampling distribution of variance example, a manufacturer measures the weights, random variables, x1 through xN, n equals 10, our sample size of 10 boxes of cereal. We know that these cereal boxes, at least the weights, are normally distributed, and they have a mean weight 12 ounces. Mu is 12. The mean is known to be 12, but the manufacturer is interested in estimating the variance of the distribution. We don't know the variance. We know the mean. And so they use that MLE for the variance, the one we just talked about. We know the sample size is 10, so we have 1 over 10. and then we sum up these 10 squared deviations around the mean, and that mean we know is 12. Now, let's suppose for a moment that the true variance was actually 0.25. We don't know the true variance. You're trying to estimate it, but let's suppose it was. Under that assumption, find the probability of the MLE is between these two values. Point two.

It might seem a little weird to suppose that the true value is something when we don't know what it is. But that light of thinking will come up again when we do hypothesis testing. That's kind of the framework. We're going to suppose some

value and say, you know, how likely is the data under that assumption? How plausible is the data under that assumption? So this is actually a common thing to do, even though it might feel a little bit weird. So suppose the true variance is actually 0.25. We want to find the probability that this MLE, which is a random variable, it's a function of random data, random variable. So we want to find the probability between 0.2. You want this probability 0.2 less than or equal to things, squared hat, then equal to 0.4. That's what we want to find. What do we know? Right. Is this a PDF? It's a random variable, so it has a distribution. In this case, it's a continuous random variable. So it would have a PDF. We're not asked to find its PDF. We're just asked to find a probability. That's all we need. If we did have a PDF, we could integrate that PDF. Yeah, sigma squared hat is a random variable. It's a statistic, so it's random. it has a distribution. In this case, it has a PDF. We're not asked to find that PDF. If we did have the PDF, we could integrate it to find a probability, because that's what you could do with PDS. But here, we just need the probability. That's all we're asked to do. No, I was going to say what piece of information do we have. But go ahead. What were you going to say? It's exactly what we're going to do. But it's a problem. Could we say that sigma square half follows a gamma distribution? Yeah, and you could get the gamma PDF and you could integrate it if you want to integrate that. Or we can try to work with our kai squared table, which might be easier. In fact, it will be easier. So we know, right, the sigma squared hat is a gamma random variable, but we're going to try to work with something kai squared, and we actually know this. We already showed this. So we're going to use that fact to know. Sigma squared hat over true variance, sigma squared, follows a kai squared random, or kai square distribution, n degrees of freedom. And in our specific case, we actually know n. We know the sample size is 10, and we're under this assumption, but the true variance is .25. So let's go ahead and just plug those in. 10 sigma squared hat over 0.25 follows kai squared distribution with 10 degrees of freedom, our sample size. What we want, this probability, 0.2, less than or equal to MLE, sigma squared hat, less than equal to 0.4. This is gamma, but we want something that's kai squared. So we're going to work with this instead. Let's just multiply all sides of this inequality by 10 over 0.25. That's all we're going to do.

It's the same. The probability 0.2 times 10 over 0.25, less than or equal to 25.

10 of sigma squared hat over 0.25, less than or equal to 0.4, 10 over 0. Multiply at all sides of the inequality by the same constant.

And we can do some simplification. This is the probability 25, that's then equal to 16.

8 less than equal to 10, sigma squared hat over 0. And we know that this thing in the middle follows a kye squared distribution. So I'm going to just say this is the probability, probability the 8 is less than equal to some random variable that I'm going to call y, it's than equal to 16, or this random variable y is kai squared with 10 degrees of freedom. And then let's just break this probability into 2.

This is the same as the probability of the random variable is less than equal to 16, minus the probability of the random variable is less than equal to 8.

So we have two left-tail probabilities, probabilities to the left of some value, for a kai squared distribution, 10 degrees of freedom. And we can look both of these up in our table. That's what our table gives us. So let's go ahead and do that. So we have 10 degrees of freedom, and we're looking at 16 and 18. Okay, so 10 degrees of freedom, this row here, and going across, I don't see 16 or 18 yet. It was 16 or 8. Across, this is the closest value to 8.8.25. That's closer to 8 than the previous one, 7.267. So I'm going to use this value, which has a probability of 0.4. That's our second probability. And then we were looking for 16 as well. So I'm going to have to go to the next page. Keep going along, look for 16, and we see 15.99, basically 16, and that probability is 0.9. Our first probability is 0.9. Our second probability was, so going back to our problem, this is roughly 0.9 minus 0.5, 1/2. There was some rounding from our discrete

approximation from our table, but that's pretty close. If it's pretty evenly between, like right between two, you can go ahead and average, but if it's closer to one, just use the one it's close to. In the real world, you'll just use software anyway. Yeah, take the average, yeah. So if it was between, you wanted something between these two, you could take the average between 0.2 and 0.25 questions here. So I have 0.4 times 10 is 4.4 divided by a quarter is 16. So we said in this middle term, we said this thing follows a kai square distribution with 10 degrees of freedom. So I just replaced it with this random variable y that's kai square with 10 degrees of freedom. So I'm not simplifying anything per se.

I'm just replacing it with the random variable. Good questions?

Okay, let's put away electronics, take out a piece of paper. We have an extra credit opportunity. If it helps, I think this is the easiest one so far. Okay, so 20 minutes starting now. You might need the table. Hopefully someone has a paper copy of the notes. No, I have the table. Oh, you have the table? I can't mean nobody brought the paper, though. Where are you always smoking? Yeah, it's true. The table. Have you? So it's just factoring at the constant. Yeah, the last, the last example. Okay, so we do what we did. Yeah.

So it's probably that's probably that's from a normal population. Because when it falls at price for... Yeah, so it's probability that's between 0.8 and 1.5. So it should just be the probability of being less than or equal to 1.5, might have the probability of less than equal 4.20. So then we just need to look up the value for 1.5 and then 0.8. And then there's 0.8 and then there's subtract. 1.5 from the value. Does the standard of surplus or factor into probably by that? Because we multiply it by that. Oh, you're right, you're right. Yeah, yeah, yeah. So we need that scalar growth. That's right, yeah. So I'm trying to figure out to go backwards. I've got that 10 over 0.25. I see the 0.25 is the length of the sigma square here, but I'm assuming it's around somewhere from here. So it's good, sigma here is 1. So that's going to be our sigma squared, right? Given the two variance in one. Oh, yeah, yeah, yeah. Okay. So it's going to be 15. So it's going to be 15 over 1. Yeah. Yeah. So then it's just going to be the thing. So then we'll have the probability. 15 times 15 times 15 times 18. 10,000, 1.5. And then, yeah. Or you're talking about it? Which is a, kind of, it sounds different. Yeah. Yeah, I have 15. That's 12. 15 times 0.8 is 12. Yeah. And the other one, the other one, and the other one, I have an extra problem. So, yeah. I'm just right now. I don't know. Oh, I mean, though, yeah. I'm trying to very simple. 22.5. There. Yeah, to be 20. There's been 20. Oh, this is really. Oh, I mean, though, yeah. So then it'll be 22. Okay, 2.5 million. And then now we look at the table with 15 degrees. And then now we look at the table with 15 degrees. They're there. Table with him, but I didn't know what. Yeah. We have to do. We're going to add. So we're going to have to. And then we have to read the freedom. And then let's see. So it's 10. This is close to different. Point three. Yeah, there we have point three. So something minus point three. Yeah. That's five. Oh, yeah. Yeah, very good. Five. Yeah. I would have gotten. I mean, just like, probably. Yeah. So then we're just closest to one. What is it? You're pointing. You're pointing. You know. You know. Yeah. Wonderful, that was probably the latest work I've ever done. Maybe the easiest, that's your question. Yeah, wait, actually. Are you thinking about this, right? So, so, yeah, yeah, that's right? Yeah, yeah, that's why I put it to 15. So basically, you take care of the moment, and you're speaking for a good. So remember how you said a high-degree with the problem of and the M-15 right? Oh, right. It's right. Yeah, right. Yeah, square. So you're thinking from the moment you should very much of those. That's right. Yeah, I think that's like ready. I heard you say I'm 12, so it's kind of rough, but that might not portray. We're not. So then we just multiply just kind of whatever. I'm sorry, no, but it's okay. And so we see. you're supposed to be right now. You are saying. So, thank you. so n times. I don't know if you're trying to cancel it out here. Why in two? Can I mean? That's what I wrote down. Because you try to cancel it out, right? You try to cancel it out? Right? So we have a problem right? Because we know that N over Sigma Square had over 602 falls at and has been right? And over the 5. 10 is your n, and 0.25% right? Right? It's closest to point in right? Well, that's an

electrical. Yeah, and then, and then, both of the probabilities by N and then dividing by sigma square. I think what you're thinking is probably the number. Well, actually, it's a number one. Oh, yeah, yeah, exactly. I just think it's a problem. Yeah. Are you going to do that? I think you're thinking about like 2.2. Yeah, that is 2. So then anyway, we end up like this, right? We can just handle for our, whatever the chosen is our significant or significant one, right? Yeah. So then it's going to be 15, um, six things right now. I want to say, honestly, I think I get on. Like, yeah. So in other words, it should be, you know, you like them. I'm like my brain. Yeah, my brain. If you're saying, if anyone, if anyone dissent it should be seen. But yeah, you know what I mean? Just to be 15. Yeah, it's a pretty good one. What's that? Ten minutes. I just need the letter. I just need the letter. And your name. Yeah. I don't know. I'm just like I'm just like looking over with it once in to make sure that. No, and for sure. Yeah, it's definitely right. It's definitely right. The only thing that I'm just somewhat I'm trying to figure out in the notes. Yeah, where does our 12 go? Because we have, is that 12? We didn't do anything yet. Yeah, yeah, what happened to it? And I don't know, at least in here, it's the true population. Yeah, we're supposing here that it's, basically. No, we're, it is the mean, but it has no relevance to what the question is asking. Like, we don't need to, we don't need to do anything with the mean to talk to me. So it's just red her right? Yeah, okay. Okay. Yeah, it's 20 or not. Oh, no. Oh. Oh, yeah. Oh, yeah. Oh. Oh. Oh. Oh, yeah. Oh, yeah. Oh, yeah. Yeah, yeah, the previous Yeah, yeah. Oh, yeah. Yeah, so you get to distribute to see like square. Oh, yeah. Oh, yeah. Yeah, yeah. Yeah, yeah. Yeah, yeah. Yeah, yeah. Oh, yeah. So, you get to see? Yeah. Do you see this? Do you do this? I think there's a lot of people that they're basically educating. Yeah, yeah, everything. I think in 2.1, this whole thing is establishing that we can trust this to be basically the norm via the C score, and then the second thing is going on. So we just follow the same pattern? Yes. We can extend that logic to make sure that it's a unit and sort of a board state that has a signal for half. It can also just be, we can put anything in a pattern. Seven minutes. Seven minutes? After class. After class. Who's still working on it? Cool. There we're again. Yeah, sorry. Thank you. Thank you. I was trying to work through this session and setting. So how are we supposed to be there? Yeah, yeah. Yeah. So how are we supposed to regard to this? Like what's the first thing we're supposed to know what's going to be about this? So I think you're just trying to prove like, like, okay, you have sigma squared which follows that distribution, or like whatever, yeah. And I think he's trying to think he was, the whole thing is trying to prove that if you multiply that by n and divide that by the true variance, that's all the kind of good distribution. But like the first thing that we're supposed to know is when we're supposing that we wanted to compose this and then you know who's going to do your answer, right? Yeah, answer. Name's good. It's easy, but I don't understand what I'm doing. I don't fully understand what I'm here. I know what you mean when you say that. I think it's just the, you know, the example was... Look at this van. He wrote the biggest C. Bro. I don't even say Roe, but do you. Is anyone else still working? Yes? I mean, he would just say simple algebra, because you just, well, he did algebra, and you're asking how to you complete this? Yeah, I suppose, like, I've just been having to be good by what I'm supposed to approach because of both these problems because it doesn't work like just normal algorithm because every single variable oftentimes represents it's self-refund. So it's very weird. So I think what's going on, right, is, so this divided by N is the formula for sigma x squared, right? So, so maybe that's right to me. So, sigma x squared formula. And this one he gave it, he gave it in the very big, like that, in the beginning of the Oh, yeah. And he'd probably believe this. Or, yeah, I know this is something that you feel like, okay, I'm telling you that the formula for is the sum of x i minus u squared over x so the estimation of it. Oh, that's it. Yeah, the estimation of the variance is equal to the sum of xi minus u squared by and the estimation here just means out there. No, estimation here means the ML. So isn't all the multiple means of reading the ML? No, I was reading one here. No, no, no, sorry. I'm not there's only one way. And I said, you know, I wasn't It would be, you find a lot. Anyone else need to turn one in? Last call. Okay, let's go over the answer now. So we have a researcher collecting a random sample, size 15. that's our sample size. Observations from a normal population. Variants, sigma squared, is unknown, but

the mean, mu, is known. It's known to be 10, and they compute the sample variance or the MLE for the variance. Some of the squared deviations around the known mean divided by the sample size 15. So this is the same exact setup as like the last example that we did in the notes. And now we're asked what's the approximate probability that this MLE for the variance falls between these two boundaries, 0.8 and 0.5, given, or under the assumption, that the true variance, sigma squared, is 1. So same setup as the last problem we did. The numbers have just changed. But we want the probability 0.8, less than are equal to sigma squared hat, than are equal to 1.5. We know that a scaled version of this, right, if we multiply this by n, the sample size, and divide by sigma squared, the true variance, that that scaled version will follow a kai squared distribution. You can take this and write this as the probability, multiply all sides of our inequality, again, by n over sigma squared, in this case, 15 over 1. 15 times 0.8 over 1, less than equal to n, sigma squared hat squared, squared, less than equal to 15 is 1.5 over 1. And simplifying 15 times 0.8, 12. The thing in the middle follows a kai square distribution, so I'm just going to write this as the random variable y. and write it as whatever you like. And that's less than equal to 22.5, this random variable that I'm calling Y, following a kai square distribution where the degrees of freedom is the sample size or 15. And we can split this probability up into two. It's just the probability of the random variables less than equal to 22.5 minus probability that the random variables less than equal to 12. And we just need to look these up in our table. If I don't have the table pulled up, I'll just give these to you now. the first one is about 0.9, and the second one is about 0.3. We would look those up the same way we did the values in the last problems in the example. And so that's equal to 0.6 or C. Any questions on this problem? It is not irrelevant. It's important because by subtracting the actual mean 10, we get something that's kye squared. Now, we didn't have to use it in any of our calculations, but by subtracting that true mean 10, that's part of what tells us that this middle thing is going to be kye squared. If we had subtracted five, well, five is not the true mean, so this whole thing would have not turned out to be kye squared. That's where it came into play. By the true mean, yeah. That's what we're going to talk about next class. We're going to consider the case where we're estimating both the mean and the variance. There will be a lot of similarities. We'll use the kai square distribution, but it'll get a little more complex. So that's actually the next couple lectures will involve that topic. It's because ultimately the kai squared relates to sum of squared standard normal random variables and standard normal is mean zero. So if x has mean that's non-zero, to get something that means zero, we have to subtract its mean, which in this case is 10.

Just the ones that turned it in, and the ones that turned it in are the ones that get the credit.

Yeah, as long as you turned in a piece of paper, you'll get the credit if enough people got it right.

50% is one point, two-thirds of the class, of the ones turning it in, is two points, and 75% is three points. No, no, it caps at three. x squared and so if you can get your random variable in that form then you can use your kye square table sorry yeah yeah exactly and this this whole concept of like taking a probability and manipulating it transforming it around to isolate random variables or get random variables that you like that's something we're going to be doing a lot of the next few weeks this is one specific case of that Yeah. Chi square, a lot of major follows, like, a normal distribution, and you kind of sample from it. So those repeated samples would follow Chi-square distribution? So part of the reason that's important is, yes, because a lot of things are normal, and it will actually relate to what we talk about next week, where we're trying to estimate both the mean and the variance of the normal distribution. Because in the real world, you're probably not going to know one or the other. You're probably going to be estimating both. And so this chi-square distribution, as we'll see, comes up in that situation, and that's part of why it's really important, and it'll come up again in hypothesis tests for various reasons.

Yeah, question. So we're still in the frequentist framework here, meaning, you know, the parameter are fixed by unknown, but we have an estimator, a statistic that we're using as an estimator. And so we want to understand the sampling distribution of that estimator, and that tells us how, you know, good or bad, in a sense, an estimator is. Does that sampling distribution have a small variance, does it have a large variance? Is it biased? Is it unbiased? And so that's kind of what we're going to be working towards is understanding the sampling distribution of our estimators for the normal distribution when we're estimating both the mean and the variance. So we don't have those tools yet. That's kind of the focus of next week right now. We're still in this case of, oh, we know the mean, we just don't know the variance, which doesn't usually come up in the real world.

But these tools will help us get there. All right, we'll have a nice weekend. And again, and I'll see you next week and we'll talk about that joint distribution.